

Some examples for computation of intersection homology

Suppose M^n is an orientable, closed, connected, n -dim mfd.

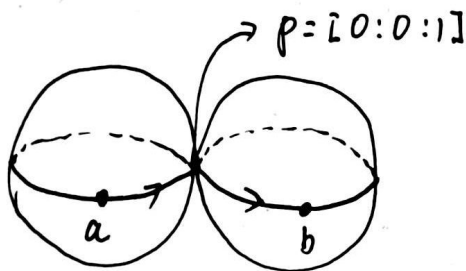
Then, $\exists$ a non-degenerate pairing with $\mathbb{Q}$ -coefficients.

$$H_i(M) \otimes H_{n-i}(M) \xrightarrow{\cap} \mathbb{Q}$$

P.D. fails for some singular spaces.

Example:

$$X = \{x, x_0 = 0\} \subset \mathbb{CP}^2 = \{[x_0 : x_1 : x_2]\}$$



$$H_0(X; \mathbb{C}) = \mathbb{C} = \langle [a] \rangle = \langle [b] \rangle$$

$$H_1(X; \mathbb{C}) = 0$$

$$H_2(X; \mathbb{C}) = \mathbb{C} \oplus \mathbb{C}$$

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P.D. for pseudo-mfd.

Let X be an oriented n -dim top pseudo-mfd. $\bar{p} + \bar{q} = \bar{b}$.

Then, $\exists$ a non-degenerate bilinear pairing.

$$IH_i^{\bar{p}}(X; \mathbb{Q}) \times I^{BM} H_{-i}^{\bar{q}}(X; \mathbb{Q}) \rightarrow \mathbb{Q}$$

Sketch of proof:

Step (a): P.D. for open cones $\mathring{c}L$.

Step (b): P.D. for $\mathring{c}L \times \mathbb{R}^k$.

Step (c): Cover X by conical neighborhood and patch local P.D. for such neighborhood by a M-V sequence.

Example: $X = \mathring{c}(\mathbb{C}P^1) \cong \mathring{c}S^2$, $X = X^3 = X^2 = X^1 \supset X^0 = \{0\}$. Consider $\bar{e}$ and $\bar{o}$.

Compute $\bar{I}H_i^{\bar{o}}(\mathring{c}(\mathbb{C}P^1))$.

$$\text{We know } \bar{I}H_i^{\bar{p}}(\mathring{c}L) \cong \begin{cases} \bar{I}H_i^{\bar{p}}(L), & i < m - \bar{p}(m+1) \\ 0, & \text{else} \end{cases}$$

$$\bar{I}H_i^{\bar{p}}(\mathring{c}(\mathbb{C}P^1); \mathbb{Q}) \cong \begin{cases} \bar{I}H_i^{\bar{p}}(\mathbb{C}P^1; \mathbb{Q}), & i < 2 - \bar{o}(m+1) = 2 \\ 0, & \text{else} \end{cases}$$

$$\bar{I}H_i^{\bar{p}}(\mathbb{C}P^1) \cong H_i(\mathbb{C}P^1) \cong \begin{cases} \mathbb{Z}, & i = 0, 2 \\ 0, & \text{else} \end{cases}$$

$$\bar{I}H_0^{\bar{o}}(\mathring{c}(\mathbb{C}P^1)) \cong \bar{I}H_0^{\bar{o}}(\mathbb{C}P^1) = \mathbb{Q}$$

$$\bar{I}H_1^{\bar{o}}(\mathring{c}(\mathbb{C}P^1)) \cong \bar{I}H_1^{\bar{o}}(\mathbb{C}P^1) = \mathbb{Q}$$

$$i \geq 2, \bar{I}H_i^{\bar{o}}(\mathring{c}(\mathbb{C}P^1)) = 0$$

$$\bar{I}H_i^{\bar{o}}(\mathring{c}(\mathbb{C}P^1)) = \begin{cases} \mathbb{Q}, & i = 0 \\ 0, & i \neq 0 \end{cases}$$

Compute $I^{BM} H_i^{\bar{e}}(c \in P')$.

$$I^{BM} H_i^{\bar{p}}(c \in P') \simeq \begin{cases} 0 & , \text{ else} \\ I H_{i-1}^{\bar{p}}(L) & , i > m - \bar{p}(m+1) \end{cases}$$

$$I^{BM} H_i^{\bar{e}}(c \in P') \simeq \begin{cases} 0 & , \text{ else} \\ I H_{i-1}^{\bar{e}}(c \in P') & , i > 2 - \bar{e}(3) = 1 \end{cases}$$

$$I^{BM} H_2^{\bar{e}}(c \in P') \simeq I H_1^{\bar{e}}(c \in P') = 0$$

$$I^{BM} H_3^{\bar{e}}(c \in P') \simeq I H_2^{\bar{e}}(c \in P') = \mathbb{Q}$$

$$I^{BM} H_i^{\bar{e}}(c \in P') = 0, i \leq 1$$

$\sum_0$

$$I^{BM} H_i^{\bar{e}}(c \in P') = \begin{cases} \mathbb{Q} & , i=3 \\ 0 & , i \neq 3 \end{cases}$$

$$I H_i^{\bar{0}}(c \in P') \times I H_{n-i}^{\bar{e}}(c \in P') \rightarrow \mathbb{Q}$$

$$i=0 \quad \mathbb{Q} \underset{\substack{[\{ \}] \\ \hookrightarrow \text{a point}}}{\downarrow}}$$

$$\mathbb{Q} \ni [\eta] \hookrightarrow \text{the whole cone} \\ \langle [\{ \}], [\eta] \rangle = 1$$

$$i=1,2,3$$

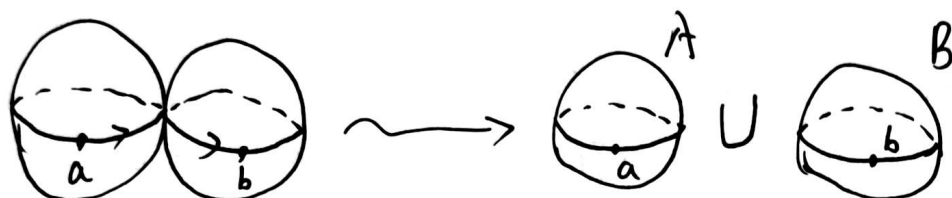
$$0$$

$$0$$

$$(3)$$

Example: $X = \{x_1, x_2 = 0\} \subset \mathbb{CP}^2$

by normalisation, $I\bar{H}_i^m(X; \mathbb{C}) \cong H_i(S^2 \vee S^2; \mathbb{C})$



$$I\bar{H}_i^m(X; \mathbb{C}) = \begin{cases} \mathbb{C} \oplus \mathbb{C} & , i=0, 2 \\ 0 & , \text{else} \end{cases}$$

As X is compact, $I^{bm} \bar{H}_i^m(X) \cong I\bar{H}_i^m(X)$.

$$I\bar{H}_0^m(X) = \begin{matrix} \mathbb{C} \oplus \mathbb{C} \\ \uparrow \quad \uparrow \\ [a] \quad [b] \end{matrix}$$

$$I\bar{H}_2^m = \begin{matrix} \mathbb{C} \oplus \mathbb{C} \\ \uparrow \quad \uparrow \\ [A] \quad [B] \end{matrix}$$

$\rightarrow [a]$ is on A

$$\langle [a], [A] \rangle = 1$$

$$\langle [a], [B] \rangle = 0$$

$\hookrightarrow a$ is not on B

$$\langle [b], [A] \rangle = 0$$

$$\langle [b], [B] \rangle = 1$$

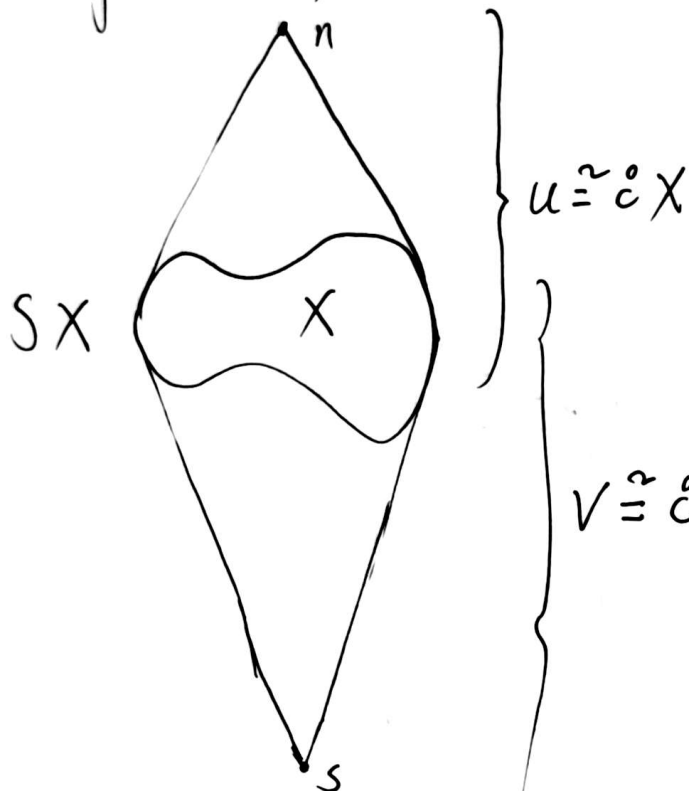
So the pairing matrix is

$$\begin{matrix} [a] & [b] \\ [A] & [B] \end{matrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

It is not degenerate.

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Example: SX , X is an n -dim compact filtered space.



$$U \cap V \cong (-1, 1) \times X \cong \mathbb{R} \times X$$

$$IH_{\star}^{\bar{p}}(\mathbb{R} \times X) \cong IH_{\star}^{\bar{p}}(X)$$

$$IH_i^{\bar{p}}(cL; \mathbb{Z}) \cong \begin{cases} IH_i^{\bar{p}}(L; \mathbb{Z}), & i < m - \bar{p}(m+1) \\ 0, & \text{else} \end{cases} \quad \text{for } L = L^m \supset L^{m-1} \supset \dots$$

$$I^{bm}H_i^{\bar{p}}(cL; \mathbb{Z}) = \begin{cases} IH_{i-1}^{\bar{p}}(L; \mathbb{Z}), & i > m - \bar{p}(m+1) \\ 0, & \text{else} \end{cases}$$

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for $i \geq n - \bar{p}(n+1)$, $IH_i^{\bar{p}}(U) = IH_i^{\bar{p}}(V) = 0$. by Mayer-Vietoris sequence,

$$\begin{array}{ccccccc} \dots \rightarrow IH_i^{\bar{p}}(U) \oplus IH_i^{\bar{p}}(V) & \rightarrow & IH_i^{\bar{p}}(SX) & \rightarrow & IH_{i-1}^{\bar{p}}(U \cap V) & \rightarrow & IH_{i-1}^{\bar{p}}(U) \oplus IH_{i-1}^{\bar{p}}(V) \rightarrow \dots \\ & \parallel \wr & & & \parallel \wr & & \\ & 0 & & & IH_{i-1}^{\bar{p}}(X) & & \mathbb{R} \times X \end{array}$$

$$\Rightarrow IH_i^{\bar{p}}(SX) \cong IH_{i-1}^{\bar{p}}(X) \quad \text{for } i \geq n - \bar{p}(n+1).$$

$$\text{If } i = 0 \geq n - \bar{p}(n+1), \quad IH_i^{\bar{p}}(SX) = 0.$$

Let $i < n - \bar{p}(n+1)$, $I\tilde{H}_i^{\bar{p}}(U \cap V) \hookrightarrow I\tilde{H}_i^{\bar{p}}(U)$ and $I\tilde{H}_i^{\bar{p}}(U \cap V) \hookrightarrow I\tilde{H}_i^{\bar{p}}(V)$ are isomorphisms.

$$\varphi: I\tilde{H}_i^{\bar{p}}(U \cap V) \hookrightarrow I\tilde{H}_i^{\bar{p}}(U) \oplus I\tilde{H}_i^{\bar{p}}(V)$$

So the $M-V$ sequence has this form

$$\cdots \rightarrow G \xrightarrow[\substack{\varphi \\ g \cdot (g, -g)}]{\substack{\varphi \\ (g, -g)}} G \oplus G \xrightarrow[\psi]{\varphi} I\tilde{H}_i^{\bar{p}}(SX) \xrightarrow[\alpha]{\varphi} I\tilde{H}_{i-1}^{\bar{p}}(U \cap V) \xrightarrow[\beta]{\varphi} I\tilde{H}_{i-1}^{\bar{p}}(U) \oplus I\tilde{H}_{i-1}^{\bar{p}}(V) \rightarrow \cdots$$

Since β is injective, $\alpha = 0$. So φ is surjective. So

$$I\tilde{H}_i^{\bar{p}}(SX) \cong \text{Im } \varphi \cong \ker \varphi \cong G \cong I\tilde{H}_i^{\bar{p}}(U \cap V) \cong I\tilde{H}_i^{\bar{p}}(X).$$

$$\text{So } I\tilde{H}_i^{\bar{p}}(SX) \cong I\tilde{H}_i^{\bar{p}}(X) \text{ for } i < n - \bar{p}(n+1).$$

Last, we compute $I\tilde{H}_{n-\bar{p}(n+1)}^{\bar{p}}(SX)$ for $n - \bar{p}(n+1) > 0$. Notice

$$\begin{array}{ccccccc} \cdots \rightarrow I\tilde{H}_{n-\bar{p}(n+1)}^{\bar{p}}(U) \oplus I\tilde{H}_{n-\bar{p}(n+1)}^{\bar{p}}(V) & \xrightarrow[\alpha]{\varphi} & I\tilde{H}_{n-\bar{p}(n+1)}^{\bar{p}}(SX) & \xrightarrow[\beta]{\varphi} & I\tilde{H}_{n-\bar{p}(n+1)-1}^{\bar{p}}(U \cap V) & & \\ & \parallel \beta & & & \downarrow \gamma & & \\ & 0 & & & I\tilde{H}_{n-\bar{p}(n+1)-1}^{\bar{p}}(U) \oplus I\tilde{H}_{n-\bar{p}(n+1)-1}^{\bar{p}}(V) & & \\ & & & & \downarrow & & \\ & & & & \vdots & & \end{array}$$

Since γ is injective, $\beta = 0$. Since $\alpha = 0$, φ is injective.

$$\text{So } I\tilde{H}_{n-\bar{p}(n+1)}^{\bar{p}} = \text{Im } \beta = 0.$$

$$\text{Hence, } I\tilde{H}_i^{\bar{p}}(SX) = \begin{cases} I\tilde{H}_{i-1}^{\bar{p}}(X), & i > n - \bar{p}(n+1) \\ 0, & i = n - \bar{p}(n+1) \\ I\tilde{H}_i^{\bar{p}}(X), & i < n - \bar{p}(n+1) \end{cases}$$

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So, if $\exists$ allowable O -chains,

$$IH_i^{\bar{p}}(SX) = \begin{cases} IH_{i-1}^{\bar{p}}(X) & , i > n - \bar{p}(n+1), i \neq 0 \\ 0 & , i = n - \bar{p}(n+1), i \neq 0 \\ \mathbb{Z} & , i = 0 \geq n - \bar{p}(n+1) \\ IH_i^{\bar{p}}(X) & , i < n - \bar{p}(n+1) \end{cases}$$

Example: $IH_i^{\bar{p}}(ST^2)$

Compute $IH_i^{\bar{0}}(ST^2)$. $n - \bar{0}(n+1) = 2 - 0 = 2$

$$0 < 2 \quad IH_0^{\bar{0}}(ST^2) = IH_0^{\bar{0}}(T^2) = \mathbb{Z}$$

$$1 < 2 \quad IH_1^{\bar{0}}(ST^2) = IH_1^{\bar{0}}(T^2) = \mathbb{Z} \oplus \mathbb{Z}$$

$$2 = 2 \quad IH_2^{\bar{0}}(ST^2) = 0$$

$$3 > 2 \quad IH_3^{\bar{0}}(ST^2) = IH_2^{\bar{0}}(T^2) = \mathbb{Z}$$

$$\text{So } IH_i^{\bar{0}}(ST^2) = \begin{cases} \mathbb{Z} & i = 0, 3 \\ \mathbb{Z} \oplus \mathbb{Z} & i = 1 \\ 0 & \text{else} \end{cases}$$

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Compute $I^{BM}H_i^{\bar{t}}(ST^2)$, $n - \bar{t}(n+1) = 2 - 1 = 1$

As ST^2 is compact, $I^{BM}H_i^{\bar{t}}(ST^2) \cong IH_i^{\bar{t}}(ST^2)$

$$0 < 1 \quad IH_0^{\bar{t}}(ST^2) = IH_0^{\bar{t}}(T^2) = \mathbb{Z}$$

$$1 = 1 \quad IH_1^{\bar{t}}(ST^2) = 0$$

$$2 > 1 \quad IH_2^{\bar{t}}(ST^2) = IH_1^{\bar{t}}(T^2) = \mathbb{Z} \oplus \mathbb{Z}$$

$$3 > 1 \quad IH_3^{\bar{t}}(ST^2) = IH_2^{\bar{t}}(T^2) = \mathbb{Z}$$

$$\text{So } I^{BM}H_i^{\bar{t}}(ST^2) = \begin{cases} \mathbb{Z} & i = 0, 3 \\ \mathbb{Z} \oplus \mathbb{Z} & i = 2 \\ 0 & \text{else} \end{cases}$$

$$H_i^{\tilde{b}}(ST^2) \times H_{n-i}^{\tilde{b}M}(ST^2) \longrightarrow \mathbb{Z}$$

$$\begin{array}{ll} i=0 & \mathbb{Z} \ni [\zeta] \quad \mathbb{Z} \ni [\eta] \\ & \hookrightarrow \text{a point in the regular part} \quad \hookrightarrow \text{the whole } ST^2 \end{array}$$

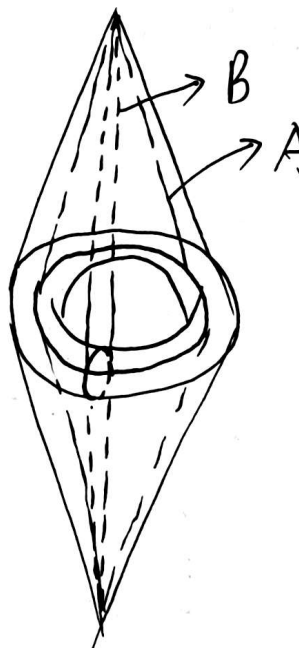
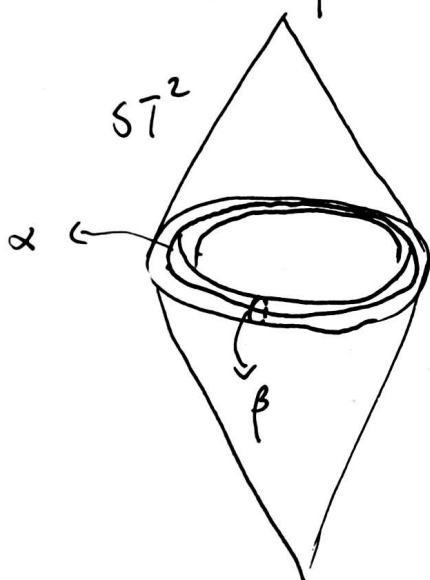
$$\begin{array}{ll} i=1 & \begin{array}{l} \mathbb{Z} \oplus \mathbb{Z} \\ \downarrow \quad \downarrow \\ [\alpha] \quad [\beta] \end{array} \quad \begin{array}{l} \text{two cycles} \\ \text{on } T^2 \end{array} \quad \begin{array}{l} \mathbb{Z} \oplus \mathbb{Z} \\ \downarrow \quad \downarrow \\ [A] \quad [B] \end{array} \quad \begin{array}{l} \text{two 2-chains} \end{array} \end{array}$$

$$i=2 \quad 0 \quad 0$$

$$\begin{array}{ll} i=3 & \mathbb{Z} \ni [\eta'] \quad \mathbb{Z} \ni [\zeta'] \\ & \hookrightarrow \text{the whole } ST^2 \quad \hookrightarrow \text{a point at regular part} \end{array}$$

$$i=1 \quad \langle [\zeta], [\eta] \rangle = 1$$

$$i=2$$



$$\begin{aligned} \langle [\alpha], [A] \rangle &= 0 \\ \langle [\alpha], [B] \rangle &= 2 \\ \langle [\beta], [A] \rangle &= 2 \\ \langle [\beta], [B] \rangle &= 0 \end{aligned}$$

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$$i=3 \quad \langle [\eta'], [\zeta'] \rangle = 1$$